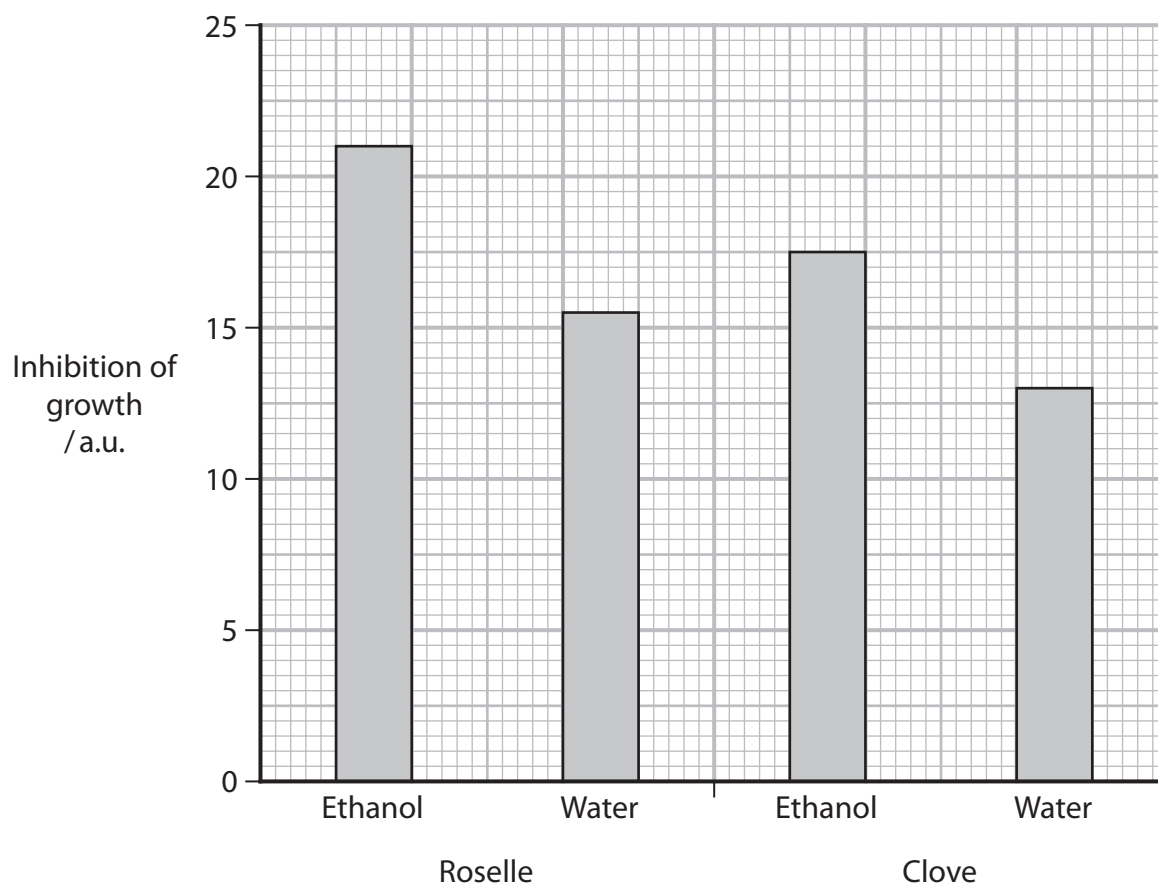


3 Scientists study the possibilities of antimicrobial properties in plants.

The effects of extracts of roselle plants and clove plants on the inhibition of the growth of the bacterium *E. coli* were studied. Extracts were made in water and in ethanol.

The results are shown in the graph.



(a) The method used in this study involved aseptic techniques.

(i) Explain **two** reasons why aseptic techniques were used in this study.

(2)

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(ii) Describe a method to obtain the data in this graph.

(4)

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(b) Draw a table to show the data in the graph.

(3)

(c) (i) Calculate the ratio of the inhibition for the ethanol extracts of roselle and clove.

(1)

Answer



(ii) Describe **two** conclusions that can be drawn from the results of this investigation.

(2)

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- (d) In another study, the effect of these extracts on the internal pH of the cells of *E. coli* was investigated.

The internal pH of *E. coli* was reduced by each of the extracts.

Suggest why these extracts have an antimicrobial effect on the bacteria.

(2)

(Total for Question 3 = 14 marks)

TOTAL FOR PAPER = 50 MARKS

